



Figure 1: Designer screen

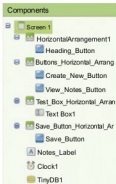


Figure 2: Components view

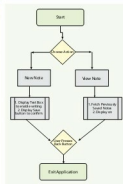


Figure 3: Block diagram

6. Make the necessary property changes as described in earlier articles, to change the text property for label and button components.

7. Renaming the components helps to identify them in the block editor.

8. So in this way, your graphical user interface is ready. Figure 1 shows what the application will look like after installation. (Figure 1)

9. Figure 2 shows the hierarchy of the components that we have dragged to the designer.

Let's look at the block diagram shown in Figure 3, so that we know what to expect and how various actions will be performed.

Now let's head towards the blocks editor to define the behaviour. Let's discuss the actual functionality that we expect from our application.

1. We need to code it so that on pressing the *New Note* button, the user is presented with a text box to write the note.
2. When the text box is enabled, we should have the *Save* button enabled at the same time so that the user can click on it and save the note.
3. On the press of the *Save* button, the data in the text box should be reflected as a note and it should also be stored in the phone database with the current time stamp.
4. On pressing the *View notes* button, it should fetch the data from the database and make it visible on the screen.
5. When the user presses the *Back* button, it should quit the application.

So let's move on and add these behaviours to the app, using the block editor. I hope you remember how to switch from the designer to the block editor. There is a button available right above the *Properties* pane to switch between the designer and block editor.

Block editor blocks

I have already prepared the blocks for you. All you need to do is to drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks the same way as in the image. I will explain what one does and how it is called.

1. The blocks below are easy to understand. The first one tells us that when the screen is initialised for the first time, there should be only two buttons visible at that time and

all the rest should be hidden.

2. When the *New Note* button is clicked, the text box should be made available so that the user can type the text and the *Save* button should also be available to click.

3. When the *Save* button is clicked, the text written in the text box should be converted into the current note and the note list should be appended with the existing record.

4. The time stamp should be noted to keep track of the note created. The clock component is able to extract this information from the system clock.

5. We are using the phone database so that when the user exits the app, the data should not be lost and for this purpose, we are using the TinyDB component.
 6. In the database, we store the data with a tag name so that when we need to fetch it back, it can be accurately received.
 7. When the *View notes* button is clicked, it should call the database component to get the data from the phone database and display it on the screen.
- Now you are done with the block editor too. Next, let's move to download and install the app on the phone to check how it works.

Packaging and testing

To test the app you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the 'Build' button. It will give you an option to download the app to your computer.
2. As the downloading begins, you can watch its progress and after it has been successfully done, the application will be placed in the download folder of your directory or the preferred location you have set for it.

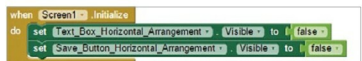


Figure 4: Block editor image 1

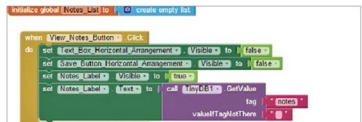


Figure 5: Block editor image 2

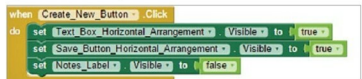


Figure 6: Block editor image 3

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